

Referee Report / Extended Commentary

Jean Tirole (2012), *Overcoming Adverse Selection: How Public Intervention Can Restore Market Functioning*, *American Economic Review* 102(1): 29–59

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1 The key idea

Tirole asks a question that the 2007–09 crisis made urgent: when a market freezes because of adverse selection, can a government *jump-start* it, and if so, how? The answer is built on a single twist relative to standard mechanism design. The textbook designer (a monopolist screening a buyer, a regulator facing a firm) takes each agent’s *reservation utility* as a primitive: the agent always has a fixed outside option, and the participation constraint is $U(\theta) \geq U_0(\theta)$ with U_0 exogenous. Tirole’s point is that when the designer is a government acting before a private market reopens, this is no longer true. The outside option of a firm that declines the government’s offer is what it can get *in the market*, and the market outcome depends on *who* declined—that is, on the mechanism itself. Reservation utilities are therefore *endogenous to the mechanism*. This two-way feedback between mechanism design and participation constraints is the paper’s conceptual contribution.

The economics that follow are sharp and somewhat counterintuitive. Because the government moves first and skims off the worst (most “toxic”) assets, it *cleans up* the residual pool and so revives the very private market it is competing against—it “creates its own competition,” exactly as Coase’s durable-good monopolist does. Anticipating the rebound, firms with good assets hold out for the market, so the government can never substitute fully for it: a *non-comprehensive* intervention with market rebound is optimal. Reviving a frozen market is necessarily expensive (the government loses money on every type it finances), yet—and this is the elegant result—the mere *existence* of the later market imposes *no* welfare cost, in contrast with the Coasian intuition that anticipated resale destroys surplus. The framework rationalizes the actual drift of TARP from asset buybacks toward equity injections, and delivers a clean public-finance logic for when a freeze should, and should not, be cured.

2 The model

2.1 Primitives

All parties are risk neutral. A *firm* (seller) is cashless, protected by limited liability, owns a *legacy asset*, and has a *new project* to finance.

Legacy asset. The asset pays R_0 on success and 0 on failure. The success probability θ is the seller's private information, drawn from a continuous c.d.f. F on $[0, 1]$ with density f . Tirole assumes F is *log-concave* (decreasing reverse hazard rate $f(\theta)/F(\theta)$), the standard regularity condition that keeps the screening problem well behaved.

New project. The project costs I . If the entrepreneur *behaves* it yields verifiable income R_1 and private benefit $b > 0$; if she *misbehaves* it yields no income but a large private benefit $B > b$ and has negative social value ($B < I$). Define the social surplus of the project

$$S = R_1 + b - I.$$

Behaving requires leaving the entrepreneur a stake $B - b$, so the *pledgeable income* is only $R_1 - (B - b)$. Three assumptions organize the analysis:

Assumption 1 (Positive NPV). $S > 0$.

Assumption 2 (Scope for credit rationing). Without the legacy asset the project cannot be financed: $R_1 - (B - b) < I$, i.e. $S - B < 0$.

Assumption 3 (Collateral may enable financing). For the best types collateral restores financing under symmetric information: $R_0 + (S - B) > 0$.

The moral-hazard wedge B is what makes the firm *credit-constrained*: it cannot raise I on the project alone and must sell (a stake in) the legacy asset, whose quality only it knows. Adverse selection then bites, exactly as in Myers and Majluf (1984).

Government and welfare. The government faces a shadow cost λ of public funds and maximizes expected gross social surplus

$$W = \mathbb{E}[U(\theta)] + \pi - (1 + \lambda)D,$$

where D is the deficit, $U(\theta)$ the seller's utility and π the buyers' profit. Since competitive buyers break even ($\pi = 0$) in equilibrium, letting $x(\theta) \in \{0, 1\}$ indicate financing,

$$W = (1 + \lambda) \mathbb{E}[\theta R_0 + x(\theta)S] - \lambda \mathbb{E}[U(\theta)].$$

The first term is efficiency (assets plus financed surplus); the second is the *social cost of the rents* the government must concede, weighted by λ . A government with $\lambda > 0$ is reluctant to leave large rents to firms—this is the tension that drives the whole design problem.

Timing and reservation utility. (1) The seller learns θ . (2) The government commits to a mechanism $C_g(\cdot)$. (3) Each seller accepts or declines. (4) Competitive buyers offer menus to the remaining sellers. (5) Sellers choose. (6) Payoffs realize. *Laissez-faire* simply deletes stages (2)–(3). For the leading “fleeting opportunity” case the autarky payoff of a seller who trades with nobody is

$$U_0(\theta) = \theta R_0.$$

2.2 The stripped-down version (buybacks only)

To isolate the mechanism, set $R_0 = 1$, suppose the entire project return accrues to the firm (no pledgeability), and take $\theta \sim U[\gamma, 1]$ with $\gamma > 0$. A seller of type θ who sells the asset at price p and finances the project obtains $p + S$; keeping the asset (autarky) is worth θ . Hence *low* types sell and high types keep:

$$\text{seller sells} \iff \theta < p + S =: \theta^*.$$

Competitive buyers break even at the *truncated mean* of what they buy:

$$p = \mathbb{E}[\theta \mid \theta < p + S] = \frac{1}{2}(\gamma + p + S) \implies p = \gamma + S, \quad \theta^* = \gamma + 2S.$$

This is Akerlof in two lines, and it delivers the discontinuity at the heart of a freeze:

- If $\gamma + S < I$ the break-even price is below I and **there is no trade**;
- If $\gamma + S \geq I$, all types $\theta < \gamma + 2S$ trade at $p = \gamma + S$.

With $S < I$ and $\gamma = 0$ the market *fully* collapses; a small piece of good news (a rise in γ) can “restart” it. Conversely, a small piece of *bad* news pushes γ below $I - S$ and freezes a previously active market—a *discontinuous* response to a marginal shock.

Adding the government. Suppose the government posts a price $p > I$. Were there no private market, all types $\theta < p + S$ would sell to it. But once those types are gone, the residual pool is better, so a private market reopens at a *higher* price—and anticipating this, the low types would rather *hold out*. The equilibrium reconciles the two: government and market prices are equated at p , types $\theta < p - S$ sell to the government, and an interval of intermediate types trades in the rebounding private market. The government loses money; private buyers break even. This is the “creates its own competition” mechanism in its barest form.

When does the government intervene? With cutoff θ^* and shadow cost λ , intervening to finance the marginal type is worthwhile only if the efficiency gain beats the extra inframarginal rents:

$$\lambda F(\theta^*) > (1 + \lambda) f(\theta^*) S \iff \lambda < 1 \quad (\text{uniform case}).$$

So on an *active* market a government with $\lambda > 1$ stays out. But if the market *freezes* (γ falls below $I - S$), it is optimal to let it rebound to its minimum volume provided

$$(1 + \lambda) S > \lambda \frac{I + S - \gamma}{2},$$

i.e. provided $I - \gamma$ is not too far above S . The same government switches from *laissez-faire* to *intervention* precisely when bad news tips the market over the freezing cliff.

2.3 The general model and its main insights

The general model relaxes uniformity, allows partial pledgeability (so the designer can take a *stake* in the project or the asset, not just post a price), and lets contracts be partial. One then works with incentive-compatible contracts and the utilities they generate: let $U_g(\theta)$ be type θ 's rent in the government mechanism and $U_m(\theta; \Theta_m)$ her rent in the market when the holdout set is $\Theta_m \subseteq \Theta$. An *endogenous-participation equilibrium* partitions types into those who join the government, Θ_g , and the market, $\Theta_m = \Theta \setminus \Theta_g$, with

$$U_g(\theta) \geq U_m(\theta; \Theta_m) \Rightarrow \theta \in \Theta_g, \quad U_m(\theta; \Theta_m) > U_g(\theta) \Rightarrow \theta \in \Theta_m.$$

The government chooses an IC mechanism, hence a rent profile $U_g(\cdot)$, to maximize W subject to the *collated* allocation being the (ideally unique) equilibrium of this fixed-point problem. The fixed point— U_g shapes Θ_m , which shapes the market rent U_m , which is the outside option in the constraint on U_g —is exactly the endogenous participation constraint.

The optimal intervention is characterized by six insights:

- (i) **Pecking order.** The government *buys back* the weakest assets outright, *recapitalizes* (leaves on the balance sheet) intermediate assets, and leaves the strongest to the market.
- (ii) **Non-comprehensive intervention / rebound.** Short of the prohibitively costly policy of buying everything, the market always rebounds; the government reduces adverse selection *enough* to let it restart, but *not too much*, to cap the fiscal cost.
- (iii) **Costly intervention.** The market already prices firms' willingness to fire-sell, so rejuvenation is necessarily expensive: the government loses money on *every* financed type.
- (iv) **No desire to shut the market down.** If the law forbids expropriation (firms get at least $U_0(\theta) = \theta R_0$), the voluntary- participation constraint can be made *costless*: the government would gain nothing from the power to ban the market, though the market deeply shapes *how* it intervenes.
- (v) **When is intervention desirable?** A market failure does not by itself justify a bailout; the budgetary cost λ makes the government reluctant. Small bad news, however, can flip the optimum from laissez-faire to intervention.
- (vi) **Intervention creates moral hazard.** The *prospect* of a bailout dulls ex-ante incentives to originate high-quality assets.

A theoretically pretty by-product (Section III) connects to the Rothschild–Stiglitz tradition: the *constrained-efficient* allocation—the one maximizing surplus subject to seller IC and buyer break-even—is an equilibrium of the laissez-faire market game, and the *unique* one surviving a “robust choice” refinement. The market, left alone, already implements the best incentive-feasible allocation; the government’s role is purely to relax the adverse-selection constraint at a fiscal cost.

3 Extensions (sketched)

Tirole gestures at several extensions; I summarize them as sketches rather than re-deriving them.

- *Ex-ante asset quality.* Endogenizing θ via costly effort confirms insight (vi): the anticipated rescue lowers the marginal return to originating good assets, so intervention breeds moral hazard. With commitment the optimal scheme would also internalize this effort margin.
- *Fungibility (Myers–Majluf).* If legacy and project payoffs cannot be told apart, contracts reward one vs. two successes; the flat bottom of the constrained optimum tilts upward but the logic is unchanged.
- *Two-sided adverse selection.* Private information on the new project too makes screening two-dimensional and hard, but under perfect correlation it collapses to one dimension and the analysis goes through.

- *A small cost of intervention* pins down a *unique* implementation: the government rescues the bottom $[\theta, \theta_g]$ and leaves $[\theta_g, \theta^*]$ to the market, maximizing the size of the free market by clearing out the worst types.

4 Placement in the course

The paper sits squarely at the intersection of the four blocks we covered, and reading it is a good way to see how they fit together.

Revelation principle and the informed-principal logic (Myerson 1982). The government’s problem is a mechanism-design problem, and the revelation principle lets us restrict attention to direct, incentive-compatible mechanisms $(U_g(\cdot), C_g(\cdot))$ —this is what licenses working with rent profiles rather than arbitrary contracts. What the paper changes is the *participation* side. In the canonical formulation the individual-rationality constraint is $U(\theta) \geq U_0(\theta)$ with U_0 fixed; here it becomes $U_g(\theta) \geq U_m(\theta; \Theta_m)$ with the right-hand side a *function of the mechanism*. The clean separation of IC and IR that we used throughout breaks down, and the designer’s problem becomes a fixed point. This is the single most important “twist on the basics” in the paper.

Adverse selection and market breakdown (Akerlof lecture). The stripped-down model *is* the Akerlof common-values market: a seller’s payoff depends on the type of whoever ends up holding the asset, the break-even price is a truncated mean, and incentive compatibility acts as a quantity/sorting constraint that generates a cross-type externality—precisely the externality that, under common values, can break decentralization (the Prescott–Townsend / Rothschild–Stiglitz theme from the Akerlof slides). Tirole’s Section III result, that the constrained-efficient allocation is the unique refined market equilibrium, is the positive counterpart to that lecture’s existence and decentralization discussion.

Competing mechanisms / multiple principals. The government and the market are two principals contracting with the same population, so the paper belongs to the competing-mechanisms world of the course (Myerson 1982 competing hierarchies; Attar–Mariotti–Salanié on non-exclusive competition for lemons). But the placement is instructive precisely through a *contrast*: that literature has the principals move *simultaneously*, which is exactly why reservation utilities there are *not* mechanism-dependent and can be treated as exogenous. Tirole’s *sequential* (Stackelberg) timing is what makes the outside option respond to the leader’s mechanism. So the paper can be read as the “competing-mechanisms” problem with the one ingredient the standard models assume away. The link to Attar–Mariotti–Salanié is doubly relevant: their result that non-exclusive competition reproduces the Akerlof outcome is what underwrites treating the rebounding market as a competitive Akerlof market in the first place.

Moral hazard (MH lecture). Adverse selection is not the whole story: the credit constraint that forces the firm to sell its asset comes from a *moral-hazard* wedge $B - b$ on the new project (Assumption 2), and the limited pledgeability $R_1 - (B - b)$ is straight out of the Holmström–Tirole “skin in the game” logic of the moral-hazard block. Insight (vi)—intervention distorting ex-ante origination—is itself a moral-hazard result layered on top of the screening problem.

5 Critical comments

These are reader’s comments rather than a referee’s line-by-line; the paper is deservedly influential and most of these are about scope.

1. **Adverse selection as *the* cause of the freeze.** The whole apparatus traces freezes to asymmetric information about asset quality. Tirole is candid (Section V) that 2007–09 also reflected fire-sale / cash-in-the-market pricing, funding runs, regulatory forbearance, counterparty risk and ambiguity aversion. A model with no leverage, no rollover risk and no interconnectedness cannot speak to the run dynamics that arguably mattered most for Lehman/Bear. The paper’s value is conditional: *if* the binding friction is adverse selection, here is the optimal response.
2. **Commitment.** The government is assumed to commit, before stage 4, to a scheme it will not renegotiate. Bailouts are famously time-inconsistent and politically renegotiated; Tirole acknowledges (Section VI) that without commitment take-up falls and sellers wait for better offers, as in the drawn-out Japanese recapitalizations. Since the no-commitment case can overturn the “costless participation constraint” result, the commitment assumption is doing real work and deserves to be flagged as such.
3. **Binary asset payoff.** The $\{R_0, 0\}$ structure is what makes the algebra so clean, but it is knife-edge: it cannot distinguish among interventions that leave the asset on the balance sheet (debt guarantees vs. equity injections). Philippon–Skreta (2012), with a continuum of returns and monotone contracts, find debt guarantees strictly optimal—a distinction Tirole’s model is, by construction, blind to.
4. **Specification of the outside option.** Several headline results, especially “no desire to shut down the market,” lean on the autarky payoff $U_0(\theta) = \theta R_0$ (fleeting opportunity) and on the no-expropriation legal constraint. With a different reservation specification—e.g. a delayed but not forgone opportunity—the costlessness of the participation constraint need not survive. The result is elegant but not robust to the timing primitive.
5. **Statics and homogeneity.** The model is static with a single cohort and a known type distribution F ; there is no aggregate uncertainty about *where* F sits, no learning, and no recovery dynamics. Yet the policy question (“is this freeze the kind that will rebound?”) is largely about that aggregate uncertainty.
6. **On the positive side—and worth saying.** The pecking-order prediction (buy the worst, recapitalize the middle, leave the best) actually rationalizes the observed drift of TARP from outright toxic-asset purchases to equity injections, and the “government must overpay” conclusion gives a coherent, if politically awkward, answer to the contemporaneous debate (Bebchuk, Krugman, Sachs vs. the “assets are undervalued, we might even profit” view). The model articulates exactly why the optimistic view was wrong: a successful purge raises residual asset prices and makes good firms hold out, so the scheme *must* lose money. That is a genuine and durable insight.

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